

Healthcare Analytics & Hospital Performance

Data Analytics Internship — Capstone Project Report

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Organization: Kinetrexa Software Private Limited

Internship Domain: Data Analytics Internship

About this report: This report documents the analysis behind a 5-page interactive Power BI dashboard (Overview, Patients, Doctors, Hospitals, Finance) built on 54,966 cleaned hospital patient records. Each section below corresponds directly to one dashboard page.

Note on the dataset: This project uses a publicly available *synthetic* healthcare dataset commonly used for analytics portfolio work. Field values are randomly generated, so relationships between variables reflect the data-generation process rather than real clinical patterns. The purpose of this project is to demonstrate a complete, professional analytics workflow — data cleaning, KPI design, multi-page dashboarding, and business communication — not to produce real medical findings.

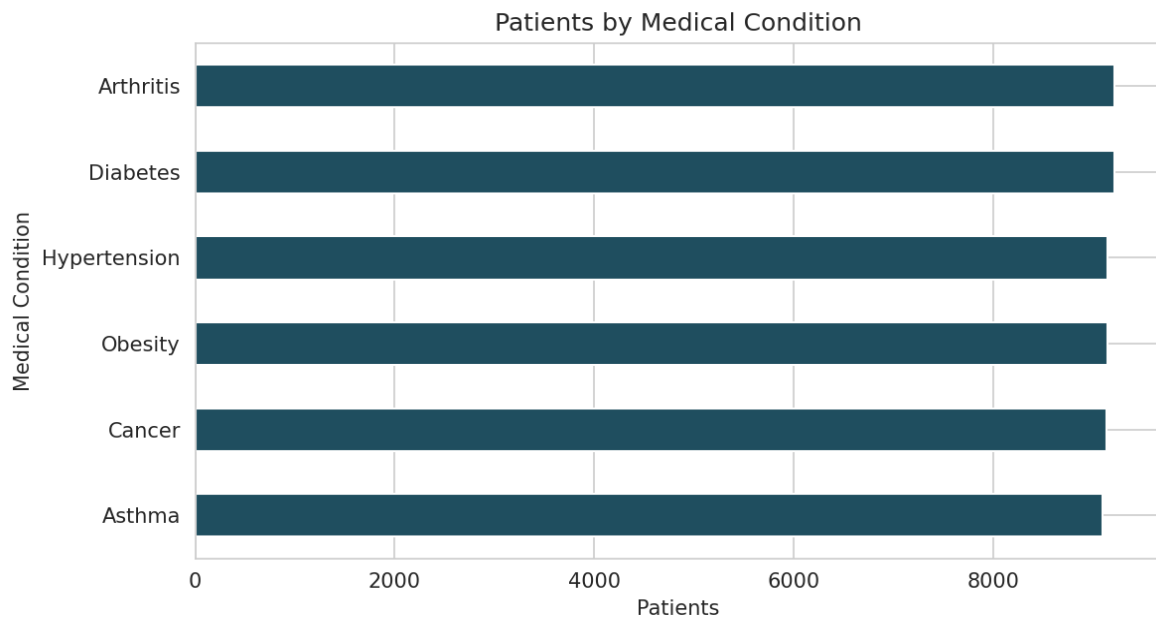
Executive Summary

Metric	Value
Total Patients	54,966
Total Revenue	\$1.40B
Average Billing per Patient	\$25,544
Average Length of Stay	15.5 days
Hospitals Represented	39,876
Doctors Represented	40,341

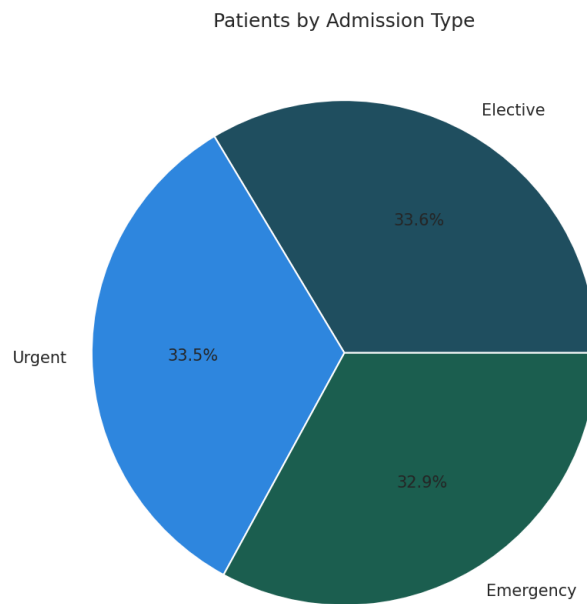
The dashboard is organized into five pages, each targeting a different stakeholder view: **Overview** (top-line KPIs for any viewer), **Patients** (demographics), **Doctors** (provider-level billing), **Hospitals** (operational metrics), and **Finance** (revenue by payer). Every page includes a written insights callout translating the charts into plain-language takeaways.

1. Overview

The Overview page gives a top-line summary: total patients, revenue, average billing, and average length of stay, alongside condition, admission type, and yearly trend breakdowns. A dynamic title and a Reset Filters button let viewers drill into a specific year and return to the full view.



Patient volume by medical condition

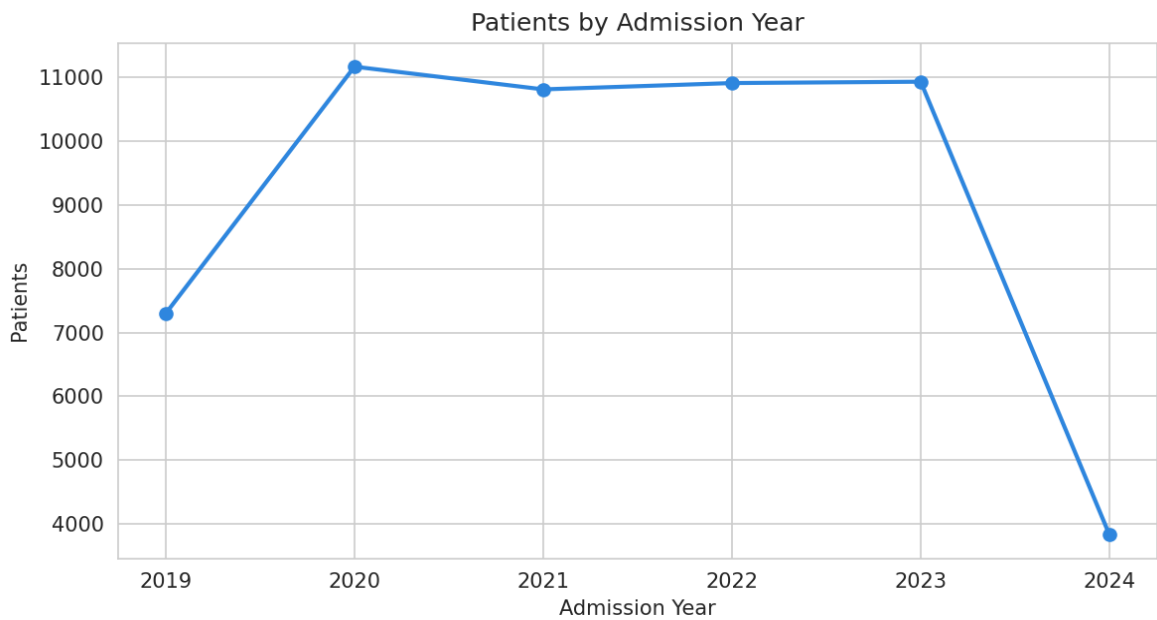


Admission type distribution

Key takeaways

- Patient volume is evenly spread across all six tracked conditions (~15-17% each).
- Admission types are nearly balanced: Elective, Urgent, and Emergency each represent roughly a third of admissions.

- Average length of stay across the full population is 15.5 days.

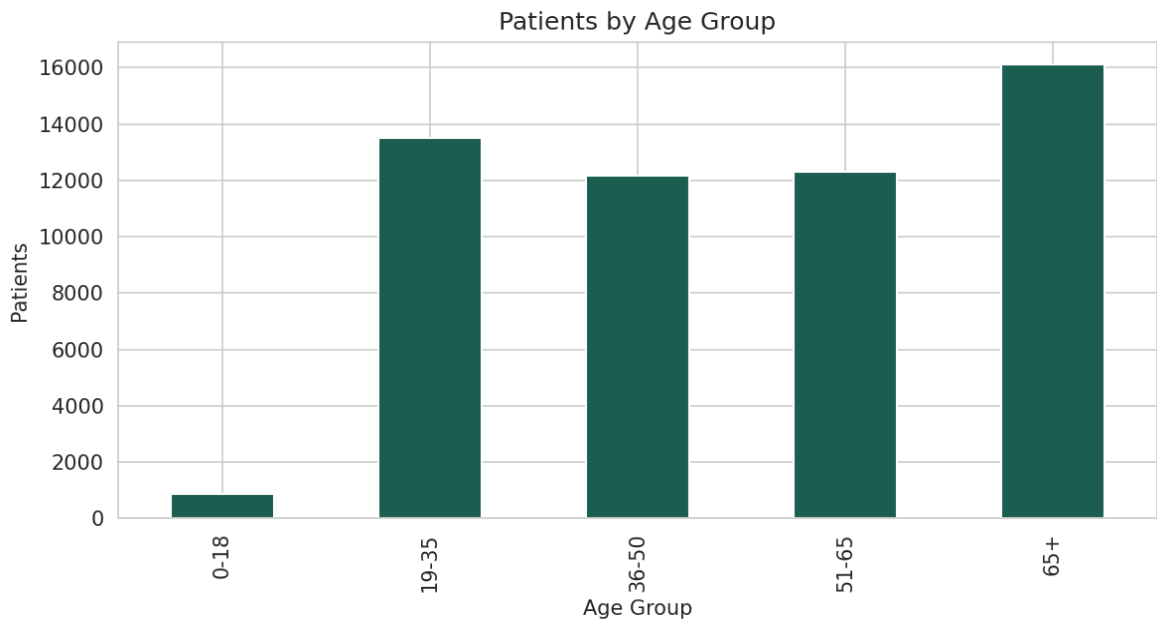


Admissions by year

Admissions were relatively stable from 2020-2023, with 2019 and 2024 reflecting partial-year data at the edges of the dataset's date range. Any trend read from this chart should account for that partial-year effect at both ends.

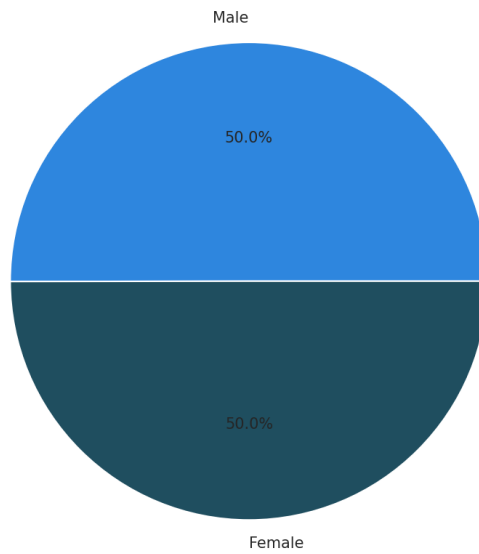
2. Patients

The Patients page profiles who is being treated — age, gender, and blood type — alongside a cross-tabulation of age group against test results to check whether outcomes skew by age.



Patients by age group

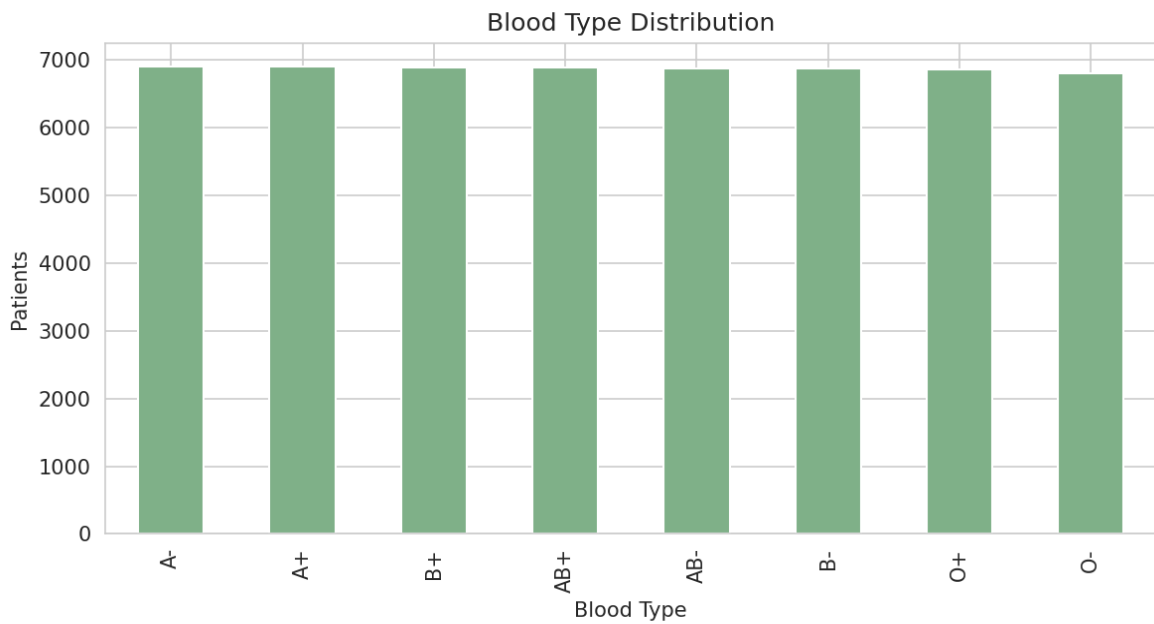
Gender Distribution



Gender distribution

Key takeaways

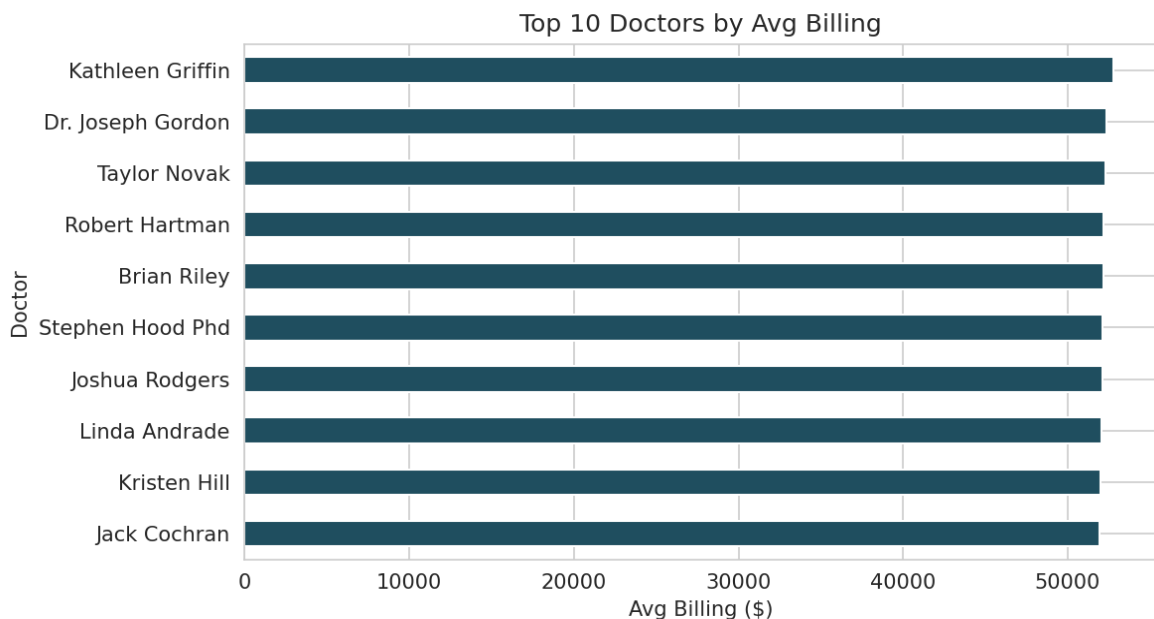
- The 65+ age group is the largest single segment (16,096 patients, ~29% of the total).
- Gender split is essentially even: 50.02% male, 49.98% female.
- Blood type distribution is roughly uniform across all eight types (~12.5% each).
- Test results (Normal/Abnormal/Inconclusive) show no meaningful skew across age groups.



Blood type distribution

3. Doctors

The Doctors page ranks providers by average billing per patient. With 40,341 unique doctors across 54,966 patients, most doctors appear only once or twice in the dataset — a direct result of how the synthetic data was generated, rather than a reflection of real caseload distribution. Because of this, the page ranks doctors by **average billing** rather than patient volume, which avoids a large tie at the top of a volume-based ranking.



Top 10 doctors by average billing

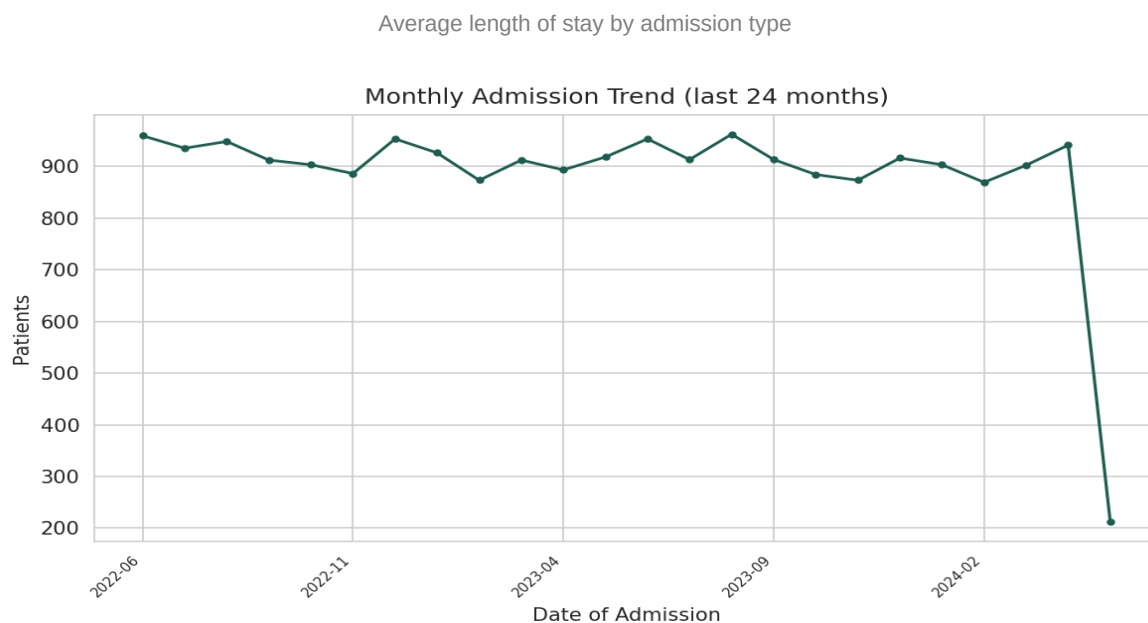
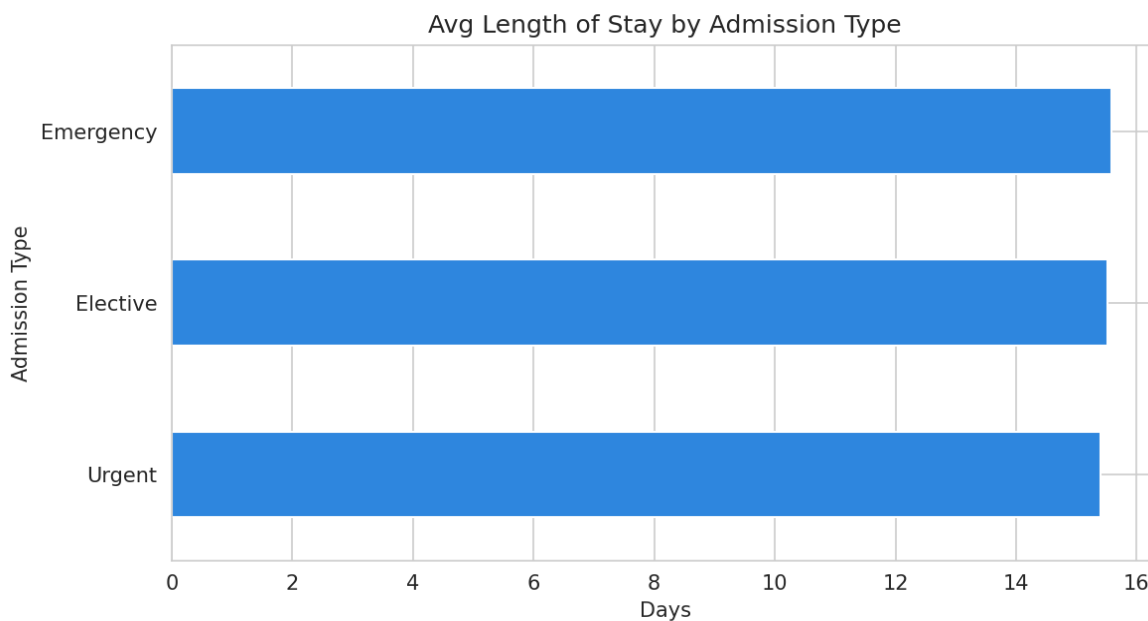
Key takeaways

- Top doctors by average billing cluster around \$52,000-\$52,800 per patient, well above the dataset-wide average of \$25,544.

- 40,341 doctors are represented across 39,876 hospitals in this dataset.
- Because most doctors have very few patients each, this page is best read as a methodology demonstration (ranking, filtering, Top-N logic) rather than a real staffing insight.

4. Hospitals

The Hospitals page focuses on operational metrics: length of stay by admission type, and admission volume trends over time.



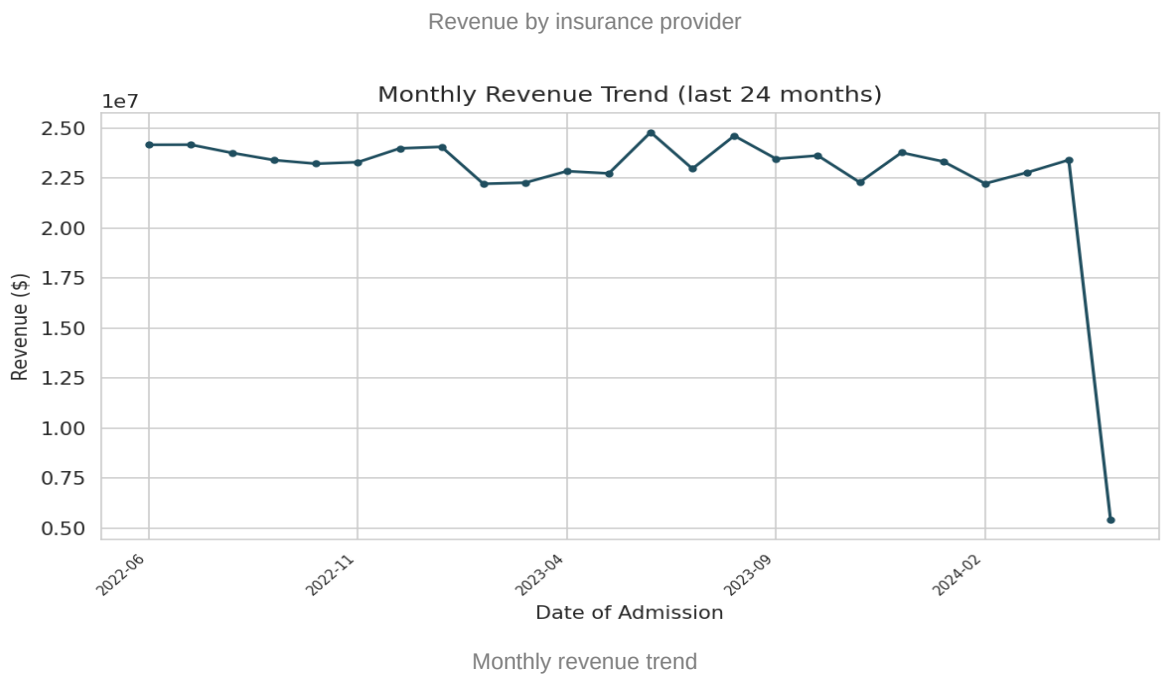
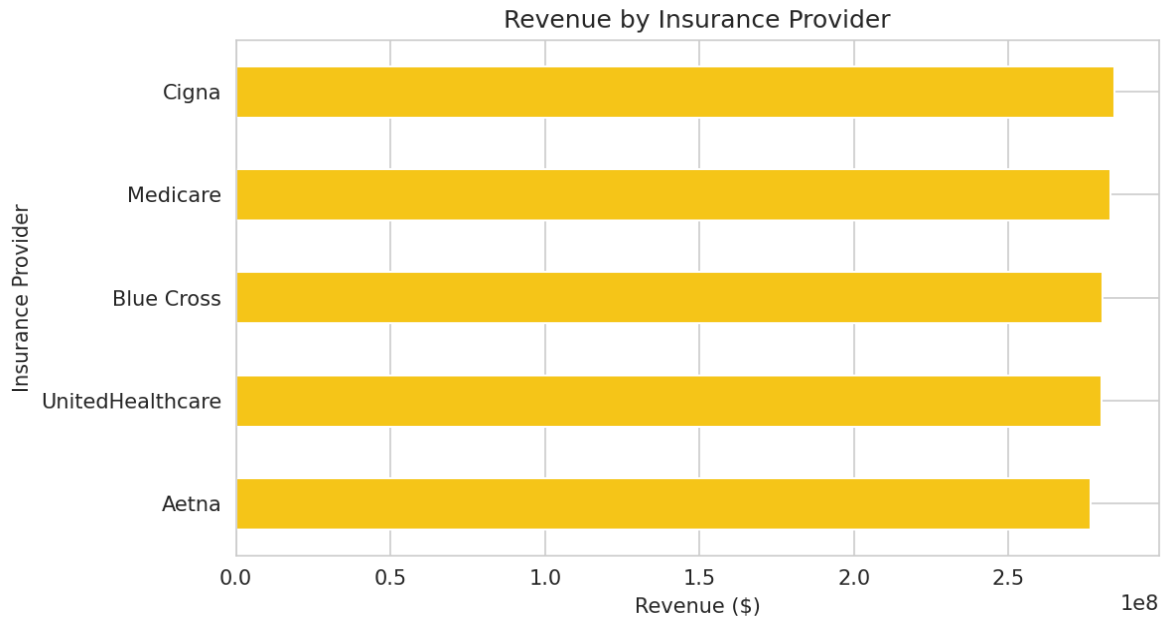
Monthly admission trend

Key takeaways

- Average length of stay is nearly identical across admission types: Emergency 15.58 days, Elective 15.51 days, Urgent 15.40 days.
- This near-uniformity indicates that, in this dataset, admission urgency does not meaningfully predict how long a patient stays — a finding worth flagging as data-generation behavior rather than a real clinical pattern.
- 39,876 hospitals are represented, most treating only 1-2 patients each given the dataset's size and structure.

5. Finance

The Finance page tracks revenue: by insurance provider and over time, giving a payer-mix and billing-trend view of the hospital system's financials.



Key takeaways

- All five insurance providers contribute a similar revenue share, ranging from \$276.5M (Aetna) to \$284.3M (Cigna) — no single payer dominates.
- Average billing per patient is consistent at \$25,544, with limited month-to-month volatility.
- Total revenue across the full dataset: \$1.40B.

Recommendations

The following recommendations are framed as if the underlying patterns were drawn from real operational data, to demonstrate how these findings would translate into action in a live hospital analytics setting:

- Track length-of-stay outliers per condition to identify discharge bottlenecks and control avoidable costs.
- Monitor revenue concentration by insurance provider to support contract negotiation leverage with payers.
- Segment admission-type analysis by condition or department (not just hospital-wide) for sharper staffing and bed-allocation decisions.
- Expand geriatric care capacity planning given the 65+ age group is the largest patient segment.
- Normalize year-over-year trend reporting to a per-month basis to avoid distortion from partial-year data at either end of the reporting window.

The full interactive dashboard, source notebook, SQL queries, and cleaned datasets are available in the accompanying GitHub repository.